The Riemann Hypothesis via Warped Stationarity

Stephen Crowley

April 22, 2026

Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Introduction

Let Z be the Hardy Z-function. The warp $\Theta(t) = \vartheta(t) + ct$ with $c > -\inf \vartheta'$ gives $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$. Y is wide-sense stationary in the Wiener time-average sense with Bochner spectrum in $[-1, 1]$. Paley–Wiener gives Y entire of exponential type ≤ 1. Akhiezer–Laguerre–Pólya places $Y \in \mathscr{LP}$. The real-line factorization $Z = \sqrt{\Theta'} \cdot (Y \circ \Theta)$ together with the Hardy Z-function axiom yields $\Re\rho = \frac{1}{2}$.

2 Axioms

Axiom 1 (Hardy Z-function). *$Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg\Gamma(\frac{1}{4} + it/2) - (t/2)\log\pi$. Z is real on $\mathbb{R}$, $|Z(t)| = |\zeta(\frac{1}{2} + it)|$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.*

Axiom 2 (Digamma asymptotic). *$\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + O(|t|^{-2}) \to +\infty$.*

Axiom 3 (Hardy–Littlewood). *$\int_0^T |\zeta(\frac{1}{2} + it)|^2\,dt = T\log(T/(2\pi)) + (2\gamma_e - 1)T + O(T^{1/2}\log T)$.*

Axiom 4 (Riemann–Siegel). *For $t > 0$, $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t)$, $N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$, $R(t) = O(t^{-1/4})$, $(d/dt)\arg R(t) = O(t^{-1/2})$.*

Axiom 5 (Wiener Thm. 29). *If $f \in L^2_{\mathrm{loc}}(\mathbb{R})$ has $c_f(\eta) = \lim_{U\to\infty} U^{-1}\int_0^U f(u)\overline{f(u+\eta)}\,du$ existing, positive-definite, continuous at 0, then for every bounded Borel $E \subset \mathbb{R}$,*

$$\mu[f](E) = \lim_{T\to\infty}\frac{2\pi}{T}\int_E \Big|\frac{1}{2\pi}\int_0^T f(u)e^{-imu}\,du\Big|^2 dm.$$

Axiom 6 (Paley–Wiener for WSS). *If $f \in L^2_{\mathrm{loc}}(\mathbb{R})$ has $\operatorname{supp}\mu[f] \subseteq [-\tau, \tau]$, then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with $|f(z)| \leq \sqrt{\mu[f](\mathbb{R})}\,e^{\tau|\Im z|}$.*

Axiom 7 (Akhiezer–Laguerre–Pólya). *If $F : \mathbb{C} \to \mathbb{C}$ is entire of exponential type $\leq \tau$, real on $\mathbb{R}$, with $\mu[F]$ finite, non-negative, supported in $[-2\tau, 2\tau]$, then $F \in \mathscr{LP}$ and every zero of F is real.*

3 Definitions

Definition 1. $\Theta(t) := \vartheta(t) + ct$, $c > c_\star := -\inf_{\mathbb{R}}\vartheta'$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := \dfrac{1}{2\pi}\displaystyle\int_0^{\Theta(T)} Y(u)e^{-i\mu u}\,du.$

4 Lemmas

Lemma 4. *$c_\star \in (0,\infty)$; for $c > c_\star$, $\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$.*

Lemma 5. *$Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t))$ for every $t \in \mathbb{R}$.*

Lemma 6. *$\vartheta''(t) > 0$ for $t > 0$.*

Lemma 7. *For $n \geq 1$ and $t \geq 2\pi n^2$, $\omega_n(t) := (\vartheta'(t) - \log n)/(\vartheta'(t) + c) \in [0, 1)$.*

5 Main Results

Theorem 8 (Time-average WSS of Y)**.** *$c[Y](\eta) := \lim_{U\to\infty} U^{-1} \int_0^U Y(u)Y(u+\eta)\, du$ exists for every $\eta \in \mathbb{R}$, is positive-definite, continuous at 0, with $c[Y](0) = 2$.*

Proof. $Y \in L^2_{\mathrm{loc}}(\mathbb{R})$: $\int_0^U Y^2\, du = \int_0^{\Theta^{-1}(U)} Z^2\, dt < \infty$. Change of variable $u = \Theta(t)$ and the Hardy–Littlewood axiom with $\Theta(T) = (T/2)\log(T/(2\pi)) + O(T)$ give $c[Y](0) = 2$. For $\eta \neq 0$, insert Riemann–Siegel into $Z(t)Z(t + \delta_\eta(t))$, $\delta_\eta(t) = \eta/\Theta'(\tau)$.

Diagonal ($m = n$, $\sigma = \sigma'$): contributes $(1/U)\int_0^{\Theta^{-1}(U)} \sum_{n\leq N(t)} n^{-1}\cos(\eta\omega_n(t))\, dt$. The frequency-ratio lemma with $\omega_n(t) \uparrow 1$ gives Cesàro convergence to $A(\eta)$ finite.

Antidiagonal ($m = n$, $\sigma = -\sigma'$): phase slope $2\vartheta'(t) + O(1) \to \infty$, with $\vartheta'' > 0$ giving monotonicity; one integration by parts per mode and summation give $o(U)$.

Off-diagonal ($m \neq n$): phase slope $\geq |\log(m/n)|$; one integration by parts per pair gives $O(1/|\log(m/n)|)$, and the Montgomery–Vaughan bound $\sum_{m\neq n\leq N}(mn)^{-1/2}|\log(m/n)|^{-1} = O(N\log N)$ with $N = O(T^{1/2})$ yields $o(U)$.

Remainder: $\int_0^T R^2 = O(T^{1/2})$; Cauchy–Schwarz against the main sum of L^2-norm $O(T^{1/2}(\log T)^{1/2})$ gives $O(T^{3/4}(\log T)^{1/2}) = o(U)$.

Positive-definiteness is inherited as a Cesàro limit of positive-definite kernels. Continuity at 0: $|c[Y](\eta) - c[Y](0)|^2 \leq c[Y](0)(c[Y](0) - \Re c[Y](\eta))$ with continuity of A at 0. □

Corollary 9. *There exists a finite non-negative Borel measure $\mu[Y]$ on $\mathbb{R}$ with $c[Y](\eta) = \int e^{i\eta\beta} d\mu[Y](\beta)$ and $\mu[Y](\mathbb{R}) = 2$.*

Theorem 10 (Uniform bound on $K[T]$)**.** *There is $C_0 > 0$ with $2\pi|K[T](\mu)|^2/\Theta(T) \leq C_0$ for every $T \geq 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \leq (2\pi)^{-2}\Theta(T)\int_0^{\Theta(T)} Y^2\, du$. Since $\int_0^{\Theta(T)} Y^2\, du = 2\Theta(T) + o(\Theta(T))$, $2\pi|K[T](\mu)|^2/\Theta(T) \leq C_0$. □

Theorem 11 (Band-limit)**.** *For $|\mu| > 1$, $\lim_{T\to\infty} 2\pi|K[T](\mu)|^2/\Theta(T) = 0$.*

Proof. Insert Riemann–Siegel into $K[T](\mu)$ and substitute $u = \Theta(t)$. The (n,σ)-mode gives $\mathcal{J}_{n,\sigma}(T,\mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u)e^{iP(u)}\, du$, $u_n = \Theta(2\pi n^2)$, $G_{n,\sigma}(u) = x^{-1/2}$, $x = \Theta'(\Theta^{-1}(u))$, $P'(u) = \sigma\omega_n(\Theta^{-1}(u)) - \mu$. The frequency-ratio lemma gives $|P'| \geq |\mu| - 1 =: \lambda > 0$. With $Q = G/P' = x^{1/2}/\hat{P}(x)$, $\hat{P}(x) = (\sigma - \mu)x - \sigma(c + \log n)$, one integration by parts and $|\hat{P}(x)| \geq \lambda x$ yield $|\mathcal{J}_{n,\sigma}(T,\mu)| \leq C_1(c, |\mu|)\beta_n^{-1/2}$, $\beta_n \geq c + \log n$. The remainder $\int_0^T R(t)\sqrt{\Theta'(t)}e^{-i\mu\Theta(t)}\, dt$ has phase slope $-\mu\Theta'(t) + O(t^{-1/2})$; the diffeomorphism lemma gives $\inf \Theta' = c - c_\star > 0$, so for t large the slope is bounded below by $|\mu|(c - c_\star)/2$, and one integration by parts gives $O(T^{-1/4})$. Summing:

$$|K[T](\mu)| \leq C_1 \sum_{n=1}^{N(T)} \frac{1}{\sqrt{n(c + \log n)}} + O(T^{-1/4}) = O\Big(\frac{T^{1/4}}{\sqrt{\log T}}\Big),$$

so $2\pi|K[T](\mu)|^2/\Theta(T) = O(T^{-1/2}/(\log T)^2) \to 0$. □

Corollary 12. $\operatorname{supp}\mu[Y] \subseteq [-1,1]$.

Proof. For every bounded Borel $E \subset \{|m| > 1\}$, the Wiener axiom gives

$$\mu[Y](E) = \lim_{T\to\infty} \int_E \frac{2\pi|K[T](m)|^2}{\Theta(T)}\, dm.$$

The band-limit theorem gives pointwise convergence to 0; the uniform bound theorem supplies the dominating function $C_0\mathbf{1}_E$; dominated convergence gives $\mu[Y](E) = 0$. Countable additivity gives $\mu[Y](\{|m| > 1\}) = 0$. $\square$

Theorem 13. *Y extends uniquely to an entire function on $\mathbb{C}$ with $|Y(z)| \le \sqrt{2}\, e^{|\Im z|}$.*

Proof. Paley–Wiener axiom with $\tau = 1$ and $\mu[Y](\mathbb{R}) = 2$. $\square$

Theorem 14. *Every zero of Y in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Y is entire of exponential type ≤ 1, real on $\mathbb{R}$, with $\operatorname{supp}\mu[Y] \subseteq [-1,1] \subseteq [-2,2]$; Akhiezer–Laguerre–Pólya with $\tau = 1$. $\square$

Theorem 15 (Riemann Hypothesis)**.** *Every nontrivial zero of ζ satisfies $\Re\rho = \frac{1}{2}$.*

Proof. By the Hardy Z-function axiom, $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$. By the factorization lemma, $Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$, so $Z(t) = 0 \iff Y(\Theta(t)) = 0$. By the preceding theorem, every zero of Y in $\mathbb{C}$ is real. Therefore every nontrivial zero of ζ lies on $\frac{1}{2} + i\mathbb{R}$, i.e. $\Re\rho = \frac{1}{2}$. $\square$